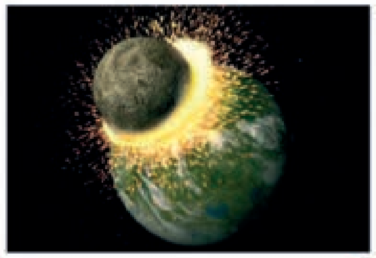
The Moon

Always in orbit around Earth, the Moon is known as Earth's satellite. It provides Earth with light during the night, though it has no light of its own – it merely reflects the Sun's light, like a mirror. It is the closest object to Earth in space, and we can see its cratered surface even with the naked eye.

HOW THE MOON FORMED

There are many theories about how the Moon came into existence. Scientists think the most likely explanation is that something collided with Earth, sending debris into space that eventually formed the Moon.

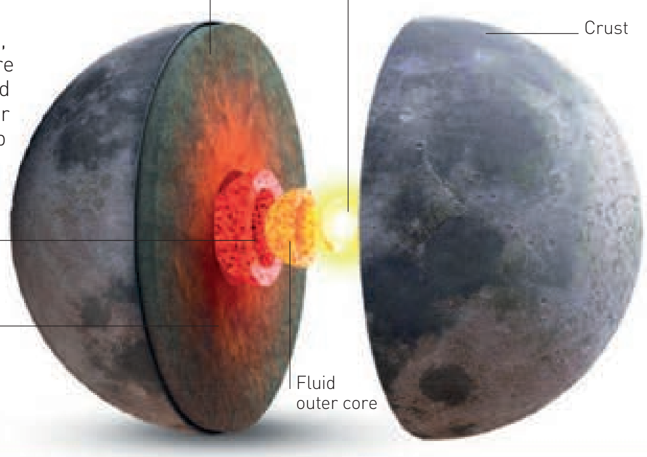


1 IMPACT A giant astronomical object hit the primitive molten Earth. The object was absorbed, but debris shot into space.

2 MOON FORMATION Earth's gravity pulled the debris into orbit, and the fragments collided and clumped together, forming the Moon.

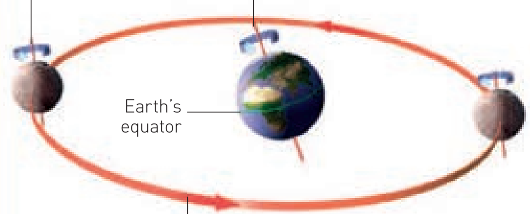
INTERNAL STRUCTURE

The Moon is made up of several layers: it has a crust, mantle, and a solid inner core surrounded by a hot and fluid outer core. There are regular "moonquakes", which last up to ten minutes.



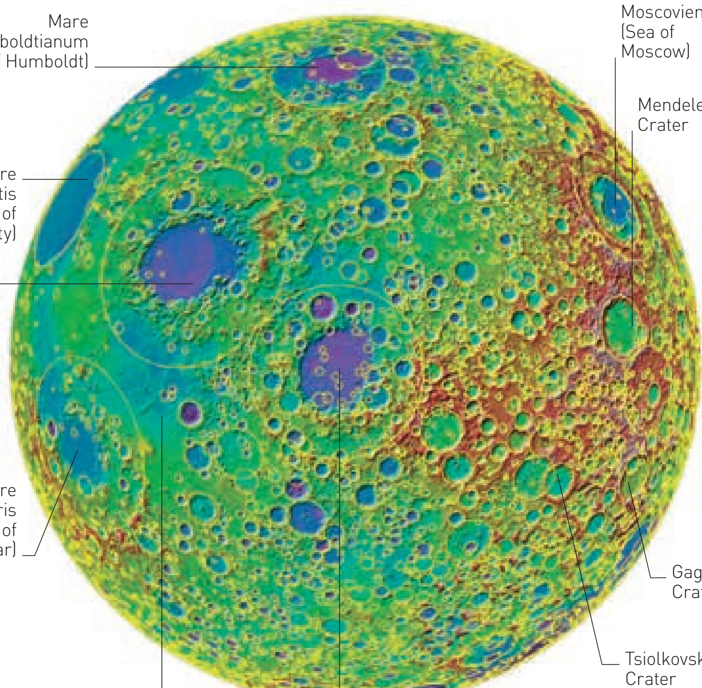
ORBITING EARTH

The Moon takes 27.3 days to orbit Earth, and the same amount of time to spin on its axis. We see some, all, or none of the Moon, depending on how much of its sunlit side faces Earth.



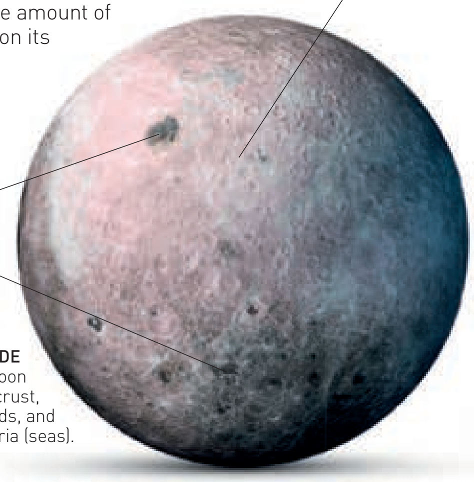
CRATERS

The Moon is rocky and pockmarked with craters formed by asteroids crashing into its surface billions of years ago. The biggest craters are called "maria", or seas. They are very flat because they were filled with volcanic lava that welled up from inside the Moon and then solidified. In this Moon map, the near side is on the left and the far side is on the right.



FAR SIDE AND NEAR SIDE

The near side of the Moon is the side that always faces Earth, because it takes the same amount of time to rotate on its axis as it does to orbit Earth.

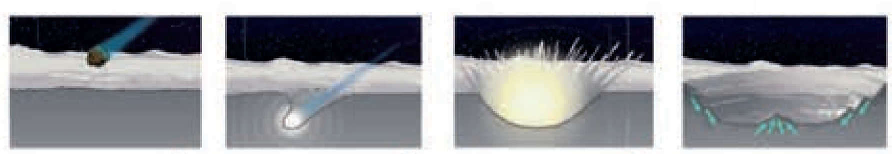


FAR SIDE This side of the Moon has a thicker crust, more highlands, and fewer maria (seas).

NEAR SIDE The near side is divided into two areas: the Lunar Highlands and maria.

HOW CRATERS FORM

When the Moon was young it was bombarded by asteroids – rocky pieces left over from the planet-making process. They blasted away the Moon's surface, forming craters, circular hollows about 10–15 times the size of the impacting asteroid.



1 INCOMING SPACE ROCK There is no atmosphere to protect the Moon from flying objects.

2 INITIAL IMPACT The object strikes the ground faster than the speed of sound, breaking the crust.

3 SHOCK WAVE On impact, the object melts and vaporizes, spewing hot rock vapour over a huge area.

4 CRATER Some of the rock vapour (ejecta flow) settles in and around the large hole that is the crater.